

Kelan S. Zielinski

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Work & Leadership Experience

Motorola Solutions Inc.

Plantation, FL

Electrical Engineering Intern

May 2024 - August 2024

- Successfully achieved optimal **impedance matching** for 109.65MHz crystal filters using a **Smith chart** to analyze and adjust circuit parameters.
- Ensured signal integrity by implementing automated tests in **LabVIEW** to verify network matching, comparing **RSSI** values to predetermined benchmarks.
- Validated scattering (**S2P**) parameters on a Vector Network Analyzer (**VNA**) and redesigned the network on Advanced Design System (**ADS**) software.
- Identified **timing issues** causing unexpected power cycles between a radio and UPS using an oscilloscope, enabling a targeted solution that improved system reliability and reduced troubleshooting time.

IoT-Driven Crop Production System

Coral Gables FL

Hardware Engineering Intern - University of Miami

January 2023 - May 2024

- Designed and built an **IoT-controlled** hydroponics system to cultivate 2000+ plants per year within a footprint of less than 50 square feet.
- Developed a hardware prototype and **validated** components, including **I2C and UART** sensors and actuators, based on system requirements.
- Utilized an **EDA** simulation platform to model and **optimize system circuitry** for improved performance and reliability.

University Of Miami – Design For Testability Laboratory

Coral Gables, FL

Teaching Assistant

January 2024 - May 2024

- Assisted students in designing and implementing a digital signal processor (**DSP**) using VHDL.
- Supported students in troubleshooting and resolving timing issues in complex digital systems, ensuring proper functionality and performance.

HKN Honor Society (IEEE) – University of Miami Chapter

Coral Gables, FL

Eta Kappa Nu – Honor Society Member

February 2024 - Present

- Supported 30+ students in mastering fundamental engineering concepts.

Projects

FPGA-Based Quantum Emulator – VHDL, Python, ModelSim

May 2025

- Designed a physics-based quantum system on an FPGA, modeling the state evolution of qubits.
- Leveraged FPGA mass-parallelism to execute **matrix exponentiation** in constant time complexity($O(1)$).
- Developed a user-friendly GUI to seamlessly communicate via **USB and UART** with the FPGA, enabling real-time control of emulation parameters and visualization of emulation results.

32-bit MIPS RISC CPU – VHDL, ModelSim, Quartus Prime

October 2024

- Designed and implemented a modular 32-bit MIPS-based RISC processor in VHDL.

Advanced Digital Signal Processor – VHDL, ModelSim

April 2024

- Implemented the ADSP-2100 digital signal processor to perform specific computations based on the instruction.
- Integrated Built-In-Self-Test (**BIST**) architecture to generate signatures and expose faults within the system design.

Rotor-based Encryption Machine – Verilog, ModelSim, Quartus Prime

November 2023

- Developed a Verilog-based model of the Enigma M3, featuring three dynamic rotors with rotational functionality.
- Verified data transfer and timing aspects of system design using **RTL** viewer and waveforms.

Education & Technical Skills

University of Miami

Coral Gables, FL

Bachelor of Science in Electrical Engineering – GPA: 3.72

May 2025

- **Relevant Coursework:** Computer Organization & Design, Digital Signal Processing(DSP), Discrete Time Signals & Systems, Communication Systems, Circuits Signals & Systems, Design for Testability(DFT), Electronics, Logic Design, Structured Digital Design, Data Structures

Technical Skills: VHDL, Verilog, C++, C, Python, ARM Assembly, Soldering, PCB Design & Testing

Developer Tools: ModelSim, ADS, Cadence, PSpice, MATLAB, Intel Quartus Prime, KiCad, Unix, LabVIEW, AutoCAD

Test Equipment: Oscilloscopes, Spectrum Analyzers, Logic Analyzers, Network Analyzers, Multimeters, Signal Generators